

FINAL ROUND • 15-SLIDE TALK

Team Huntrix

Presentation Script

Evidence-first selective inference for Bengali hallucination detection

Delivery target

Approximately 14 minutes, 35 seconds across four presenters. Rehearse to about 14 minutes to preserve pause and handoff time.

Touhidul Alam Seyam • MD. Abtahee Kabir • Joyeta Barua Moni • Noore Tamanna Orny

Presenter-ready rehearsal edition | 23 July 2026

Run of show

Presenter	Slides	Time	Primary arc
Touhidul Alam Seyam	1-4	3 min 35 sec	Problem, hypothesis, router
MD. Abtahee Kabir	5-8	4 min 5 sec	Structured solvers and judge
Joyeta Barua Moni	9-12	4 min 5 sec	Evidence, experiments, reproducibility
Noore Tamanna Orny	13-15	2 min 50 sec	Phase 2 runtime, shift, close

Speaking rule

Do not race the diagrams. State the claim first, then point to the mechanism or number on screen. Finish the final sentence before handing off.

Technical wording to keep exact

The frozen notebook uses one local llama.cpp server whose model is tensor-split equally across two T4 GPUs.

PRESENTER BLOCK

Touhidul Alam Seyam

Slide 1: Evidence-first selective inference for Bengali hallucination detection

Presenter Touhidul Alam Seyam **Target** 45 seconds

Good morning. We are Team Huntrix. This talk is about a research question rather than a leaderboard position: when is a Bengali hallucination decision exact enough for a deterministic method, and when should the system abstain to a language-model judge? We will show how that question changed our architecture, which experiments failed, how the decision log shaped later versions, and what the organizer-run Phase 2 notebook revealed about source shift. The final score appears later as one evaluation view, alongside local tests, reproducibility checks, and runtime evidence.

Slide 2: One binary label hid many different reasoning tasks

Presenter Touhidul Alam Seyam **Target** 50 seconds

The label looked simple: zero for hallucinated and one for faithful. The rows were not simple. Some asked whether a response followed a supplied passage. Others needed arithmetic, option parsing, Bengali morphology, idiom meaning, legal sections, or a historical fact. One generic judge prompt had to switch between all of these without being told which evidence standard to use. It gave us a useful baseline, but it also created a ceiling. A response could sound reasonable while answering the wrong relation. That pushed us toward task routing.

Slide 3: The working hypothesis became a selective cascade

Presenter Touhidul Alam Seyam **Target** 55 seconds

Our fix was a selective cascade. We normalize the row, identify its task family, and try only the rules that belong to that family. The technical contribution is the combination of selective routing, small task solvers, claim-level verification, typed provenance, and an offline two-T4 fallback. Exact lineage can decide an exact source match. Structured solvers can decide a parsed number or option. Context checks can decide a contradiction inside the supplied passage. Each stage can also return nothing. If the source is only similar, or the relation is unclear, the row reaches the judge. The next slide opens this router and shows the actual branch conditions.

Slide 4: Routing made abstention an explicit state

Presenter **Touhidul Alam Seyam** Target **65 seconds**

This is the router in operational form. After normalization, the first question is whether the row contains an answer-bearing context. If it does, the context verifier compares atomic claims. Without context, the router checks for a structured task such as a number, range, option, or morphology problem. The next branch asks whether an exact source lineage exists. Even then, typed evidence must match the entity, predicate, and value. A conflict between sources of equal trust forces abstention. Every branch has the same contract: return zero, one, or null. Null is not an error. It sends the row forward. Only the final unresolved tail reaches the LLM judge.

Handoff

Abtahee will open the structured and relation-verification paths.

PRESENTER BLOCK

MD. Abtahee Kabir

Slide 5: Structured solvers shared one decision contract

Presenter MD. Abtahee Kabir **Target** 60 seconds

The deterministic branch was not one large collection of keywords. It contained separate parsers with the same output contract. The numeric solver normalized Bengali digits, units, dates, and ranges before checking the exact comparison requested. The multiple-choice solver extracted the option set, mapped aliases, and verified that the response selected a valid option. The morphology solver identified the requested form and compared the root, suffix, or grammatical class. Idiom and lexical rows first distinguished literal from figurative meaning. All four produced an entity, predicate, parsed value, source, and verdict. If any required field was missing or contradictory, the solver returned null. This shared contract made the router testable even though the parsers were task-specific.

Slide 6: A correct fact can still answer the wrong relation

Presenter MD. Abtahee Kabir **Target** 65 seconds

This released example shows why semantic similarity was not enough. The context says that Tareque Masud's latest feature film was Runway and that it was released in 2010. The prompt asks when the film was released. The response says Runway. That is a correct title and the correct entity, but it has the wrong semantic type. The expected predicate is release year, while the response supplies a film title. A retrieval system may see a highly related answer. Our relation gate rejects it and emits label zero. This is the concrete failure mode that motivated the predicate field used in later slides.

Slide 7: Context rows were reduced to claim-level entailment tests

Presenter MD. Abtahee Kabir **Target** 65 seconds

This is the expanded view of Figure 2a in the report. For context rows, raw similarity was not the decision variable. We split the context into clauses and the response into atomic factual claims. Alignment used anchors such as named entities, numbers, negation, and the requested relation. Each claim then received one of three states: supported, contradicted, or unknown. A single contradiction was enough for label zero. Label one required support for every factual claim and no unsupported addition. Anything between those cases produced abstention and moved to the judge. This claim-level rule matters when most of a response copies the context but changes one year, person, quantity, or polarity.

Slide 8: Qwen judged 158 rows, not the whole test set

Presenter MD. Abtahee Kabir Target 55 seconds

The fallback used Qwen3.6 27B in a 16.332 GiB quantized checkpoint. Context rows received one entailment view. No-context rows received up to three short views: a factual check, strict exam grading, and a check that was deliberately cautious about faithful labels. A no-context response needed unanimous positive votes to receive label one. The first zero ended evaluation early. We disabled thinking and constrained generation to one binary token. Row order, prompt order, seeds, and batch boundaries were fixed. That made the judge fast enough to be a tail component rather than the whole system.

Handoff

Joyeta will show how accepted evidence was traced, then connect that design to the experiments and final result.

PRESENTER BLOCK

Joyeta Barua Moni

Slide 9: A source record was accepted only after four fields aligned

Presenter Joyeta Barua Moni Target 65 seconds

This is the expanded view of Figure 2b in the report. Every deterministic fact was stored as a typed evidence record. The record carried a source identifier, source family, normalized entity, predicate, value, and provenance. At prediction time, four gates had to pass. The entity had to match. The requested predicate had to match, not merely appear near the same topic. The value needed exact or task-specific equivalence. Finally, no source of equal trust could contradict it. The resulting trace recorded the route, source, predicate, and verdict. If any gate failed, the verdict was null and the row moved forward. This trace made error analysis possible without storing a manual label vector for the test set.

Slide 10: Versioned comparisons locate the main design pivots

Presenter Joyeta Barua Moni Target 60 seconds

This table makes the evolution easier to compare, but it is not a controlled ablation because each version accumulates more than one change. The baseline scored 0.657. A uniform Qwen judge raised that to 0.797 but judged every row under one policy. v6 introduced source and structured routing and reached 0.903, the largest observed step. v37 tightened provenance and reduced the unresolved queue to 312 rows. v85 added a reproducible two-source anchor and left 193 rows. v91 required typed idiom and predicate guards, reached 0.967 publicly, and left 158 rows. The important evidence is the alignment between score movement, queue reduction, and documented architectural changes, not a claim that each row is a pure single-variable experiment.

Slide 11: Decision logs turned failed runs into routing rules

Presenter Joyeta Barua Moni Target 60 seconds

The decision log mattered because a failed experiment had to change the next experiment. Prompt voting looked reasonable on the released set and scored 0.783 publicly, so we stopped treating prompt variation as independent evidence. The TF-IDF verifier reached 0.826 out of fold but missed cultural facts, so we retained it only as a portable fallback. Raw retrieval often found the right entity and the wrong relation, so retrieval was restricted to proposing candidates. v87 changed two labels using plausible facts and reduced the public score from 0.966 to 0.965. We reverted it and added the source-plus-predicate admission rule. Later changes also had to preserve the regression suite and emit an evidence trace.

Slide 12: Reproducibility was treated as an evaluation result

Presenter Joyeta Barua Moni **Target** 60 seconds

We treated reproducibility as a result, not an administrative detail. The codebase had 249 automated tests, including 225 deterministic controls. The notebook reads the organizer's literal test path, loads pinned model, runtime, and evidence assets, and uses a fixed row, prompt, and batch schedule. One local llama.cpp server tensor-splits the model equally across the two T4 GPUs. The final validator preserves incoming IDs and writes exactly id and label. Three full runs produced byte-identical files, the frozen hash matched the scored CSV, and the organizer-run notebook validated 5,000 Phase 2 rows in 3 hours and 52 minutes. These checks establish execution integrity; they do not substitute for an unreleased Phase 2 accuracy score.

Handoff

Orny will cover the measured runtime and Phase 2 source shift.

PRESENTER BLOCK

Noore Tamanna Orny

Slide 13: Phase 2 completed in 3 h 52 min with 95.9% judged

Presenter Noore Tamanna Orny Target 65 seconds

The organizer-run Phase 2 notebook processed exactly 5,000 rows on two Tesla T4 GPUs. The router finished in 73.18 seconds but resolved only 207 rows, or 4.14 percent. The remaining 4,793 rows went to Qwen. The model was ready 171.25 seconds after the notebook started, and the judge stage took 13,747.50 seconds. End-to-end runtime was 13,921.91 seconds, which is 3 hours, 52 minutes, and 1.91 seconds. The output validation cell completed successfully with 5,000 ordered IDs and binary labels. This used 42.97 percent of the nine-hour budget. It is measured runtime evidence, not a Phase 2 accuracy score.

Slide 14: Phase 2 source shift collapsed deterministic coverage

Presenter Noore Tamanna Orny Target 60 seconds

Phase 2 changed the source distribution as expected. The organizer listed Common Crawl, recent newspapers, Wikipedia, Banglapedia, government service pages, Bangladesh law, NCTB textbooks, and literature. Deterministic coverage fell from 93.72 percent in Phase 1 to 4.14 percent in the held-out run. The router did not compensate by forcing approximate evidence. It abstained, and 95.86 percent of rows reached the fixed judge. That raised runtime to 3 hours and 52 minutes, but the notebook still finished with more than five hours of headroom. Accuracy remains an open question until the organizer provides a score.

Slide 15: What the experiments support—and what remains open

Presenter Noore Tamanna Orny Target 45 seconds

We will close by separating supported findings from open questions. The experiments support three design choices: task families benefit from separate policies, predicate alignment reduces relation swaps, and explicit abstention keeps the pipeline safe when deterministic coverage falls. The main limitations are equally important. We had only 299 released labels, aggregate competition scores provide no row-level errors, and the Phase 2 accuracy result is still unknown. When those labels or analyses become available, the next work is to inspect the judged tail, recalibrate uncertain cases, and expand licensed evidence for civic and news domains. What we can already verify is reproducibility and runtime compliance. We are ready for your questions.

Likely questions after the talk

Q1. Why did you not fine-tune a classifier or the judge?

Suggested answer: The released set had only 299 labeled rows, while Phase 2 deliberately changes the source distribution. Fine-tuning on that small set would make it easy to learn Phase 1 artifacts. Routing gave us a larger gain, preserved an interpretable abstention path, and kept inference inside the Kaggle limit.

Q2. Why Qwen3.6-27B?

Suggested answer: It is an open multilingual model that handled Bengali factual and entailment prompts well in our local checks. The 16.332 GiB Q4_K_P checkpoint fits the two-T4 environment, and the short constrained output keeps the fallback practical.

Q3. Did exact-source matching amount to manually labeling the test set?

Suggested answer: No. We did not manually assign labels to test rows, use hidden labels, or call paid or closed APIs. The rules apply typed relations to public or competition-permitted evidence. A source match is allowed to decide only when the entity, value, and requested predicate agree; otherwise the system abstains.

Q4. Did the judge process 158 requests?

Suggested answer: It processed 158 unique rows. Context rows use one view. Some no-context rows use two or three short views, with early exit on the first zero, so the request count is higher than the row count.

Q5. How does the system handle Phase 2 sources it has never seen?

Suggested answer: Unknown sources do not receive approximate deterministic labels. Context checks and general structured solvers still run, but lineage rules abstain unless their exact conditions are met. The unresolved row then reaches the fixed LLM judge.

Q6. Why are the public and private scores different?

Suggested answer: The public leaderboard evaluated only part of the test set. The private score came from the remaining held-out portion. We report both aggregate scores, but we do not infer row-level labels or error patterns from the leaderboard.

Source map

Use

These numbers match the compact citations shown in the deck. Full bibliographic details remain in the technical report.

1. Bengali LLM Hallucination Detection Challenge, data description, leaderboard, and rules:
<<https://www.kaggle.com/competitions/bengali-hallucination>>
2. Organizer Phase 2 source preview: <<https://www.kaggle.com/competitions/bengali-hallucination/discussion/719835>>
3. *BenHalluEval: A Multi-Task Hallucination Evaluation Framework for Large Language Models on Bengali*:
<<https://arxiv.org/abs/2605.31483>>
4. *When Words Don't Mean What They Say: Figurative Understanding in Bengali Idioms*:
<<https://arxiv.org/abs/2602.12921>>
5. Qwen3.6-27B model card: <<https://huggingface.co/Qwen/Qwen3.6-27B-FP8>>
6. llama.cpp: <<https://github.com/ggml-org/llama.cpp>>
7. Team Huntrix experiment log, frozen manifest, deterministic trace, Kaggle runtime records, and final leaderboard records in this repository.
8. Organizer-executed Phase 2 notebook `huntrix-1.ipynb` and its final runtime report: 5,000 rows on Tesla T4 x2, completed July 22, 2026.

Final delivery cue

Finish on the open questions and runtime evidence, pause, look at the judges, and say: "We are ready for your questions."